

Planning for the Next Pandemic

As the world becomes more connected, the risk of major pandemics is increasing. (Madhav et al.) Unprepared governments and health systems have suffered the debilitating effects of the HIV, H1N1, and most recently COVID pandemics in just the past 50 years; and the next pandemic may be on our doorstep. As of late July 2024, there are 6 confirmed human cases of H5N1, a highly pathogenic avian influenza (Garg). Even if H5N1 does not cause a pandemic, another infectious disease will likely do so soon, demonstrating the need for preparation. Though infectious diseases cannot always be prevented, steps must be taken to limit the loss of life and disruption to society for the next pandemic.

With the loss and suffering from COVID-19 fresh on our minds, we will be tempted to focus on the COVID pandemic's specific failings: for example, on stockpiling PPE and increasing hospital capacity, among other things. However, due to the variable nature of infectious diseases, the next pandemic could look radically different from COVID. Thus, it is necessary to consider what long-term investments we can make that will be broadly applicable to preventing many types of pandemics. To this end, this paper will zoom in on medical mistrust, infrastructure improvements, and facilitating global cooperation. We will make broad, salient recommendations based on the analysis of multiple historic pandemics: COVID-19, smallpox and cholera.

Firstly, steps must be taken to foster trust in medical institutions. Medical mistrust decreases the uptake of public health tools, which are vital in managing pandemics, such as vaccinations. Consider the case of COVID-19: Allen et al. finds that each increase of 1 point on

the Medical Mistrust Index is associated with a 16% decrease in the odds of vaccination in their survey. Elucidating the reasons for this medical mistrust is important for repairing it. Historically, research on medical mistrust has focused on Black and Native populations, but a growing body of literature shows the lack of trust in communities with a lower socioeconomic status, the LGBTQ+ community, rural communities, and more. Some elements that fuel medical mistrust include historical abuses (such as the Tuskegee Study or the campaign to introduce opioids as “safe”) as well as ongoing discrimination and mistreatment (Allen et al.) (Lister and Joudrey). Efforts must be made to invest in research on these phenomena and to remedy them, as a first step to forging trusting relationships between communities and healthcare facilities.

Another way to build trust is to bridge the gap between the clinic and the community. Many healthcare institutions have partnered with community members— often chosen by focus groups— who maintain amicable, trusting relationships with the individuals in their community (Nandyal et al.). Integration within the community establishes the healthcare system as an institution working for and with the community, not against it. For example, Yale’s Center for Clinical Investigation recruits “cultural ambassadors” who are figures of authority in community organizations— for example, Priest Leroy Perry at a majority Black church in Connecticut— who help plan and participate in events designed to increase participation in clinical trials, and act as a representative for their community by assisting in the development of protocols for trials. This approach promotes community connection and gives the community members more input in clinical trials (*Cultural Ambassadors*). Clinical Trials can occupy an especially “scary” part of the scientific world, as they evoke fears of being a lab rat and being experimented on, but this cultural ambassador program helps assuage these worries and give a window into clinical trials from a trusted source, which ultimately increases participation in Yale’s trials. Trust cannot be

created as a crisis emerges and fear runs wild- it is important to build meaningful and trusting connections long before the next pandemic. Connecting directly with the community and building up trusting partnerships is a valuable mechanism for building trust in medical institutions from the public, which will ultimately allow for higher uptake of vital public health measures by the public in the next pandemic.

Investment in various types of infrastructure is also vital, such as maintaining sustainable farming and affordable housing, because these structures can fuel the rapid spread of infectious diseases. Consider the cholera outbreaks in 19th-century London, where a major factor fueling the outbreak was the contaminated water supply. In the 1840s, human excrement was discarded in the same cesspools where drinking water was obtained, and by 1854, pump water was drawn from the similarly unhygienic Thames, which had excrement and a milieu of other contaminants that contained *Vibrio cholera* (Johnson 11, 105). This case underscores the necessity for well-developed food and water infrastructure. The United States has some of the safest and cleanest water in the world, so water and sewage infrastructure is less relevant. However, other infrastructure, such as our agriculture system, may trigger future pandemics. Policies that prioritize sustainable farming may reduce the likelihood of zoonotic spillover. Most pandemics are zoonotic, meaning they come from animals and then pass onto humans (Madhav et al.). Growing livestock demand leads to crowded livestock conditions and habitat loss. The former may increase the chances of a disease spreading rapidly amongst animals and then spilling over to humans. The latter increases contact between humans and wildlife, increasing the odds of a spillover event(*Intensive Farming*). Promoting more sustainable farming should mitigate these risks. Indeed, a One-health approach reveals the interconnectedness of animal and human health. Thus, actions taken to create more sustainable agriculture and maintain proper conditions for

livestock will have ripple effects on human health (*International Cooperation for the Future of Pandemic Preparation* | *Wilson Center*). Alongside sustainable agriculture, affordable and accessible housing should be invested in. This will limit overcrowded living situations in which respiratory diseases are spread quickly. (“Household Crowding”). Housing and agricultural structures, as well as other types of infrastructure that promote disease transmission, should be re-evaluated in order to limit the propagation of infections.

Finally, the facilitation of global cooperation in terms of research, medical treatments, and data sharing is paramount. Due to increasing levels of globalization, what occurs in one corner of the earth will invariably impact the rest of it, both in positive ways and negative ways, so the US cannot only focus on what occurs within our own borders. On the detrimental side of things, COVID was recognized in China by December and in the US by January, demonstrating the need for coordinated responses to prevent the spread of a future pandemic (*CDC Museum COVID-19 Timeline* | *David J. Sencer CDC Museum* | *CDC*). On a more beneficial side, countries can benefit from crosstalk about treatments and research. For example, Lady Mary Wortley introduced variolation in the UK after learning about it in her time in Turkey. She was not the first westerner to notice: she noted that other expatriates partook in variolation, but seemingly made no attempt to translate this finding to the West (Williams 84). Earlier knowledge sharing could have saved many from dying of smallpox. The government should incentivize global research by providing funding for knowledge-sharing interventions and seek to create global partnerships with other health and science institutions.

The government should also make an effort to promote seamless data sharing among countries. As in the case of the cholera epidemic of 1854 at the Broad Street Pump, Snow was unable to make his conclusions without a plethora of statistics, including about where residents

got their water. Farr made a critical addition by adding the pump used on his weekly mortality bill, which allowed Snow to decipher the mystery behind cholera transmission and help to set up germ theory (Williams 102). Modern pandemics are just as in need of data for the promotion of medical interventions and the advancement of science.

Though this analysis is not exhaustive, medical mistrust, infrastructure, and global cooperation are all springboards from which the United States can begin to prepare for the next pandemic. Analysis from past pandemics shows the breadth of experience and necessary factors in reducing mortality and morbidity, as well as societal upheaval during a pandemic. Though these suggestions would not typically be viable in the short-minded Congress and political system of the United States, because of their long-term commitment and initial outlay, the benefits in mitigating the next pandemic justify the immensity of these proposals. The United States must take bold and firm action now, for the next pandemic may be a way off, but the work needed to minimize its impact begins now.

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